Songning Lai

Machine Learning Quantitative Research Engineer · Trustworthy AI / CTA Research

lais0328eee@gmail.com · xll0328.github.io · Google Scholar 886 / h-index 13 / i10-index 20

SL

PUBLIC EDITION

PUBLIC EDITION · SCOPE SHOWN, IMPLEMENTATION WITHHELD

Role, dates, and medium-to-high-frequency CTA research scope are shown; employer names, systems, datasets, metrics, and implementation details are withheld.

ROLE	PUBLIC SCOPE	RESEARCH
ML quantitative research	End-to-end CTA research	Trustworthy AI / CBMs

ROLE & TIMELINE

Sep 2026 - Present

Machine Learning Quantitative Research Engineer

Large quantitative research organization

End-to-end medium-to-high-frequency CTA research; work-specific details withheld.

Jan 2026 - Aug 2026

Quantitative Research / AI Algorithms

Quantitative research organization

Quantitative research and AI algorithm development; project details withheld.

Nov 2025 - Jan 2026

Quantitative Research / AI Algorithms Intern

Public employer details withheld

Quantitative research and AI algorithm exploration.

EDUCATION

Sep 2020 - Jun 2024

Shandong University

B.Eng. in Communication Engineering · School of Information Science and Engineering / Chongxin Honors Program

RESEARCH AREAS

Trustworthy AI; concept bottleneck models; editable and continual learning; multimodal interpretation; medical AI; reliable time series; robustness; LLM interpretability.

RESEARCH PROFILE

Machine learning quantitative research and trustworthy AI researcher. Public professional scope: end-to-end medium-to-high-frequency CTA research; academic work covers concept bottleneck models, reliable time series, multimodal interpretation, and robustness.

PUBLIC ACHIEVEMENTS

ICRA 2026 WBCD Logistics Picking World 1st; WWW 2026 Workshop TIME Best Paper Runner-Up; CISP-BMEI 2022 Best Paper; NeurIPS 2024 and ICRA 2025 Travel Awards; National Mathematical Modeling First Prize (Top 0.6%); MathorCup National First Prize (Top 3%).

RESEARCH SNAPSHOT

19 first-, co-first-, or corresponding-author papers; Google Scholar: 886 citations, h-index 13, i10-index 20.

SERVICE & TOOLS

Reviewer for ICLR, NeurIPS, ICML, CVPR, ICCV, EMNLP, KDD, AAAI, IJCAI, ACM MM, and related venues. Tools: Python, PyTorch, pandas, NumPy, scikit-learn, Git/Linux.

PRIVACY BOUNDARY

Role, dates, and research scope are public. Employer names and work-specific implementations remain private.

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PUBLIC EDITION

Selected public publications

1. NUM2EVENT: Structured Event Inference from Numerical Time-Series
EMNLP 2026 Findings · Co-first author

2. Dynamic-V2C: Editable and Continual Vision-to-Concept Bottleneck Models via Influence Functions
ECCV 2026 · First author

3. SANT-CBM: Structurally-Aware and Noise-Tolerant Semi-supervised Concept Bottleneck Models
MICCAI 2026 · Corresponding author

4. Concept-Consistent Semi-Supervised Learning for Concept Bottleneck Models via Confidence-Guided Pseudo-Label Propagation
ICANN 2026 · Corresponding author

5. Multimodal Deception in Explainable AI: Concept-Level Backdoor Attacks on Concept Bottleneck Models
TMLR 2026 · First author

6. ACE: Attribution-Controlled Knowledge Editing for Multi-hop Factual Recall
ICLR 2026 · Co-first author

7. Towards Reliable Time Series Forecasting under Future Uncertainty: Ambiguity and Novelty Rejection Mechanisms
ICASSP 2026 · Co-first author

8. Learning New Concepts, Remembering the Old: Continual Learning for Multimodal Concept Bottleneck Models
ACM MM 2025 · First author

9. From Guesswork to Guarantee: Towards Faithful Multimedia Web Forecasting with TimeSieve
ACM MM 2025 · First author

10. DRIVE: Dependable Robust Interpretable Visionary Ensemble Framework in Autonomous Driving
ICRA 2025 · First author

11. Stable Vision Concept Transformers for Medical Diagnosis
ECML-PKDD 2025 · Co-first author

12. PEPL: Precision-Enhanced Pseudo-Labeling for Fine-Grained Image Classification in Semi-Supervised Learning
ICASSP 2025 · Co-first author

13. Faithful Vision-Language Interpretation via Concept Bottleneck Models
ICLR 2024 · First author

14. Towards Multi-dimensional Explanation Alignment for Medical Classification
NeurIPS 2024 · Co-first author

15. Shared and Private Information Learning in Multimodal Sentiment Analysis with Deep Modal Alignment and Self-supervised Multi-task Learning
IJCNN 2024 · First author

16. A Comprehensive Review of Community Detection in Graphs
Neurocomputing 2024 · Co-first author

17. Multimodal Sentiment Analysis: A Survey
Displays 2023 · First author

18. Cross-domain Car Detection Model with Integrated Convolutional Block Attention Mechanism
Image and Vision Computing 2023 · Co-first author

19. Predicting Lysine Phosphoglycerylation Sites using Bidirectional Encoder Representations with Transformers & Protein Feature Extraction and Selection
CISP-BMEI 2022 · First author

Selected coauthored publications

1. ClinStab: Stability-Oriented Learning for Medical Time Series via Dual-Stream Alignment
NeurIPS 2026 Poster · Coauthor

2. Mimicking the Physicist's Eye: A VLM-centric Approach for Physics Formula Discovery
NeurIPS 2026 Poster · Second author

3. Decomposing Visual Histories with Vision-Language Agents: Hierarchical Temporal Guidance for Compositional Image Generation
EMNLP 2026 Main Conference · Coauthor

4. Demystifying Hidden-State Recurrence: Switchable Latent Reasoning with On-Policy Reinforcement Learning
EMNLP 2026 Findings · Coauthor